Jiaxuan Li

Personal Information

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Research Interests

- Dwarf galaxies across different environments: satellite galaxies, field dwarfs, ultra-diffuse (puffy) galaxies.
- Dark matter physics: observational tests for dark matter models.
- Low surface brightness astronomy: galaxy outskirts, tidal debris, data reduction, and instrumentation.
- Stellar population: star formation history, surface brightness fluctuation, initial mass function, photo-z.
- Galaxy evolution: quenching, scaling relations, galaxy–halo connection.
- Statistical methods and machine learning in astrophysics.

Research Positions

Brinson Prize Fellow, KIPAC, Stanford University	Sept 2026 – Aug 2029
KIPAC Fellow, KIPAC, Stanford University	Sept 2029 – Aug 2031
Research Assistant, KIAA, Peking University, China	Sept 2020 – Aug 2021

Education

Doctor of Philosophy, Department of Astrophysical Sciences, Princeton University — Aug 2021 – July 2026

Advisor: Jenny E. Greene

Thesis: *Probing Small-Scale Structure and Galaxy Formation with Nearby Dwarf Galaxies*

Master of Arts, Department of Astrophysical Sciences, Princeton University — Aug 2021 – May 2023

Bachelor of Science (highest honor), Peking University, China — Sept 2016 – July 2020

Major: Astrophysics GPA: 3.80/4.00 Rank: 2 / 28

Advisors: Pieter van Dokkum (Yale) & Luis C. Ho (Peking)

Press Coverage

- AAS Nova: Satellite Surplus Around Dwarf Galaxy DDO 161
- AAS Nova: Discovery of a Lonely Galactic Hedgehog
- The New York Times: Scientists Say They've Found a Dwarf Planet Very Far From the Sun
- Reuters: Possible new dwarf planet spotted near the edge of the solar system
- Sky & Telescope: Another Dwarf Planet in Our Solar System?
- Princeton University: Princeton Astronomers Discover Extraordinary Distant Object at Solar System's Edge

Honors and Awards

Brinson Prize Postdoctoral Fellowship	Feb 2026
KIPAC Postdoctoral Fellowship, Stanford University	Feb 2026
American Astronomical Society's Rodger Doxsey Travel Prize Honorary Winner	Jan 2026
Charlotte Elizabeth Procter Fellowship, Princeton University	Apr 2025
The fellowship recognizes final-year graduate students for outstanding academic performance and professional promise.	
American Astronomical Society's Chambliss Astronomy Achievement Student Award	Jan 2024
Awarded to the top graduate student posters at the AAS meeting.	
Outstanding Undergraduate Thesis Award in Beijing (北京市本科优秀毕业论文)	Sept 2020
Inaugural PKU Junior Scholar ("未名学士" 称号), Peking University	June 2020
Outstanding Graduate of Universities in Beijing (北京市普通高校优秀毕业生)	June 2020
Outstanding Graduate of Peking University (北京大学优秀毕业生)	June 2020
PKU Scholar in Physics (未名物理学子)	2017 – 2020
Tang Li-Xin Scholarship (10,000 RMB per year, most competitive scholarship in PKU)	May 2019
AEON Scholarship, Peking University (10,000 RMB, 2/202)	Sept 2018
Leo KoGuan Scholarship, Peking University (10,000 RMB, 4/202)	Oct 2017
Lin-bridge Prize for Excellent Undergraduate Research (2,800 RMB, endowed by Douglas Lin)	Sept 2018
Merit Student, Peking University	2017, 2018
First Prize, 8th China Undergraduate Physicists Tournament	Aug 2017
Meritorious Winner in Mathematical Contest In Modeling (MCM/ICM)	Apr 2018
Silver Medal, 9th International Olympiad on Astronomy and Astrophysics (IOAA)	Aug 2015
Gold Medal & Best Result, China National Astronomy Olympiad	2014, 2015
Gold Medal (3rd place), 1st Princeton University Physics Competition	Jan 2015

Publications

Total citations: 572; first-author citations: 145; h-index: 16 ADS Library

1. **Li J.**, Greene J. E., Danieli S., Carlsten S. G., Moustakas J. et al., ELVES-Dwarf. II. A Systematic Search for Satellite Systems of Dwarf Galaxies in the Local Volume, arXiv:2609.00283.
2. **Li J.**, Greene J. E., Danieli S., Carlsten S. G., Geha M. et al., ELVES-Dwarf. I. Satellite Systems of Eight Isolated Dwarf Galaxies in the Local Volume, *ApJ* 1002, 75 (2026).
3. **Li J.**, Greene J. E., Danieli S., Carlsten S. G., Geha M., A Possible "Too-many-satellites" Problem in the Isolated Dwarf Galaxy DDO 161, *ApJL* 998, L24 (2026).
4. **Li J.**, Greene J. E., Carlsten S. G., Danieli S., Hedgehog: An Isolated Quiescent Dwarf Galaxy at 2.4 Mpc, *ApJL* 975, L23 (2024).
5. **Li J.**, Melchior P., Hahn C., Huang S., PopSED: Population-level Inference for Galaxy Properties from Broadband Photometry with Neural Density Estimation, *AJ* 167, 16 (2024).
6. **Li J.**, Greene J. E., Greco J. P., Huang S., Melchior P. et al., Beyond Ultra-diffuse Galaxies. I. Mass-Size Outliers among the Satellites of Milky Way Analogs, *ApJ* 955, 1 (2023).

7. **Li J.**, Greene J. E., Greco J., Beaton R., Danieli S. et al., Beyond Ultra-diffuse Galaxies. II. Environmental Quenching of Mass-Size Outliers among the Satellites of Milky Way Analogs, *ApJ* 955, 2 (2023).
8. **Li J.**, Huang S., Leauthaud A., Moustakas J., Danieli S. et al., Reaching for the Edge I: probing the outskirts of massive galaxies with HSC, DECaLS, SDSS, and Dragonfly, *MNRAS* 515, 5335 (2022).
9. Mandacarú Guerra L. S., **Li J.**, Pan Y., Greene J. E., Danieli S. et al., The Merian Survey: A Statistical Census of Bright Satellites of Sub-Milky Way-Mass Galaxies, arXiv:2610.03981.
10. Carlsten S. G., **Li J.**, Greene J. E., Drlica-Wagner A., Danieli S., A Sample of Nearby Isolated Dwarf Galaxies: A First Look at the Mass Function of Field Dwarfs, *ApJ* 1001, 244 (2026).
11. Carlsten S. G., **Li J.**, Greene J. E., Drlica-Wagner A., Danieli S., ELVES-Field: Isolated Dwarf Galaxy Quenched Fractions Rise below $M_* \approx 10^7\ M_\odot$, *ApJL* 1001, L42 (2026).
12. Cheng S., **Li J.**, Yang E., Discovery of a Dwarf Planet Candidate in an Extremely Wide Orbit: 2017 OF201, *ApJL* 998, L6 (2026).
13. Wei L., Huang S., **Li J.**, Sun Z., Li M. et al., Zangetsu: A Candidate Isolated, Quiescent, and Backsplash Ultra-diffuse Galaxy in the COSMOS Field, *ApJ* 997, 32 (2026).
14. Nemer A., Hahn C., **Li J.**, Melchior P., Goodman J., Constraining Protoplanetary Disk Winds from Forbidden Line Profiles with Simulation-based Inference, *ApJ* 965, 157 (2024).
15. Ou X., Necib L., Cuesta-Lazaro C., Torrey P., Ahvazi N. et al. (including **Li J.**), From starlight to dark matter: a stochastic interpolation approach to map dark matter from stellar density, arXiv:2609.27028.
16. Mintz A., Greene J. E., Danieli S., Kado-Fong E., Leauthaud A. et al. (including **Li J.**), Galaxies Below the Fundamental Metallicity Relation Are Morphologically Disturbed: Merian H Morphologies and DESI Metallicities, arXiv:2609.26887.
17. Le M. N., Knapen J. H., Ivezić ., Junais, Watkins A. et al. (including **Li J.**), Rubin LSST DP2 unveils almost-dark galaxies in the Virgo Cluster, arXiv:2609.19246.
18. Rose J. C., Lisanti M., Torrey P., Navarro F. V., Garcia A. M. et al. (including **Li J.**), The DREAMS Project: Disentangling the Impact of Halo-to-halo Variance and Baryonic Feedback on Milky Way Satellite Galaxies, *ApJ* 1006, 244 (2026).
19. Lin X., Fan X., Cai Z., Liu Y., Sun F. et al. (including **Li J.**), $(\mathrm{LRDs})^2$: The Low-ReDshift Little Red Dots Survey. II. DESI DR1 Sample, arXiv:2605.21574.
20. Garcia A. M., Rose J. C., Torrey P., Caputo A., Lisanti M. et al. (including **Li J.**), The DREAMS Project: Disentangling the Impact of Halo-to-halo Variance and Baryonic Feedback on Milky Way Dark Matter Density Profiles, *ApJ* 1002, 8 (2026).
21. Lilie E., Rose J. C., Lisanti M., Garcia A. M., Torrey P. et al. (including **Li J.**), The DREAMS Project: Disentangling the Impact of Halo-to-Halo Variance and Baryonic Feedback on Milky Way Dark Matter Speed Distributions, *ApJ* 1002, 168 (2026).
22. Ma Y., Greene J. E., Setton D. J., Goulding A. D., Annunziatella M. et al. (including **Li J.**), Counting Little Red Dots at z < 4 with Ground-based Surveys and Spectroscopic Follow-up, *ApJ* 1000, 59 (2026).

23. Pan Y., Danieli S., Greene J. E., **Li J.**, Leauthaud A. et al., The Merian Survey: A Statistical Census of Bright Satellites of Milky Way Analogs, *ApJ* 1000, 119 (2026).
24. Zimmerman E. A., Gal-Yam A., Groot P. J., Ofek E. O., van Roestel J. et al. (including **Li J.**), A Faint Progenitor System for the Faint Supernova 2024vjm, arXiv:2602.09096.
25. Danieli S., Kado-Fong E., Huang S., Luo Y., Li T. S. et al. (including **Li J.**), First Data Release of the Merian Survey: A Wide-field Imaging Survey of Dwarf Galaxies at $z \sim 0.06$─0.10, *ApJ* 993, 110 (2025).
26. Luo Y., Wick J., Leauthaud A., Wetzel A., Jones T. et al. (including **Li J.**), Testing the Stellar Feedback-driven Breathing Mode in Low-mass Galaxies with Gas Kinematics, arXiv:2510.17996.
27. Zeng Z. C., Peter A. H., Du X., Benson A., **Li J.** et al., Diversity and universality: Evolution of dwarf galaxies with self-interacting dark matter, *Phys. Rev. D* 112, 063008 (2025).
28. Srinivasaragavan G. P., Hamidani H., Schroeder G., Sarin N., Ho A. Y. et al. (including **Li J.**), EP250108a/SN 2025kg: A Jet-driven Stellar Explosion Interacting with Circumstellar Material, *ApJL* 988, L60 (2025).
29. Mintz A., Greene J. E., Kado-Fong E., Danieli S., **Li J.** et al., A Nonparametric Morphological Analysis of Hα Emission in Bright Dwarfs Using the Merian Survey, *ApJ* 974, 273 (2024).
30. Dou J., Peng Y., Gu Q., Renzini A., Ho L. C. et al. (including **Li J.**), The H I Reservoir in Central Spiral Galaxies and the Implied Star Formation Process, *ApJL* 973, L23 (2024).
31. Luo Y., Leauthaud A., Greene J., Huang S., Kado-Fong E. et al. (including **Li J.**), The Merian survey: design, construction, and characterization of a filter set optimized to find dwarf galaxies and measure their dark matter halo properties with weak lensing, *MNRAS* 530, 4988 (2024).
32. Greene J. E., Danieli S., Carlsten S., Beaton R., Jiang F. et al. (including **Li J.**), ELVES. III. Environmental Quenching by Milky Way-mass Hosts, *ApJ* 949, 94 (2023).
33. Keim M. A., van Dokkum P., Danieli S., Lokhorst D., **Li J.** et al., Tidal Distortions in NGC1052-DF2 and NGC1052-DF4: Independent Evidence for a Lack of Dark Matter, *ApJ* 935, 160 (2022).
34. Greene J. E., Greco J. P., Goulding A. D., Huang S., Kado-Fong E. et al. (including **Li J.**), The Nature of Low-surface-brightness Galaxies in the Hyper Suprime-Cam Survey, *ApJ* 933, 150 (2022).
35. Shi J., Peng Y., Diemer B., Stevens A. R., Pillepich A. et al. (including **Li J.**), Cold Gas in Massive Galaxies as a Critical Test of Black Hole Feedback Models, *ApJ* 927, 189 (2022).
36. Danieli S., van Dokkum P., Trujillo-Gomez S., Kruijssen J. D., Romanowsky A. J. et al. (including **Li J.**), NGC 5846-UDG1: A Galaxy Formed Mostly by Star Formation in Massive, Extremely Dense Clumps of Gas, *ApJL* 927, L28 (2022).
37. Liu Q., Abraham R., Gilhuly C., van Dokkum P., Martin P. G. et al. (including **Li J.**), A Method to Characterize the Wide-angle Point-Spread Function of Astronomical Images, *ApJ* 925, 219 (2022).
38. Miller T. B., van Dokkum P., Danieli S., **Li J.**, Abraham R. et al., The Dragonfly Wide Field Survey. II. Accurate Total Luminosities and Colors of Nearby Massive Galaxies and Implications for the Galaxy Stellar-mass Function, *ApJ* 909, 74 (2021).

39. van Dokkum P., Lokhorst D., Danieli S., **Li J.**, Merritt A. et al., Multi-resolution Filtering: An Empirical Method for Isolating Faint, Extended Emission in Dragonfly Data and Other Low Resolution Images, *PASP* 132, 074503 (2020).
40. Danieli S., Lokhorst D., Zhang J., Merritt A., van Dokkum P. et al. (including **Li J.**), The Dragonfly Wide Field Survey. I. Telescope, Survey Design, and Data Characterization, *ApJ* 894, 119 (2020).
41. Han J. J., Chiti A., Chen K., Bechtol K., Bellini A. et al. (including **Li J.**), A Path to an All-Sky Survey with Roman, white paper for the Roman Cycle 1 observation, arXiv:2602.21280.
42. **Li J.**, Greene J., Danieli S., Carlsten S., Geha M., ELVES-Dwarf Survey: Probing Satellite Population of Isolated Dwarf Galaxies in the Local Volume, AAS abstract (2026).
43. **Li J.**, Carlsten S., Danieli S., Greene J. E., Lin X. et al., Testing the Mass Threshold of Reionization-Quenching with Isolated Dwarf Galaxy Hedgehog, HST Proposal (2025).
44. Cheng S., **Li J.**, 2017 OF201, Minor Planet Electronic Circulars (2025).
45. **Li J.**, Melchior P., Hahn C., Huang S., PopSED: Population-Level Inference for Galaxy Properties from Broadband Photometry with Neural Density Estimation, AAS abstract (2024).
46. Greene J. E., Mapping the 3D structure of the nearby Universe with Roman+Surface Brightness Fluctuations, NASA Roman Proposal (2024).
47. **Li J.**, `smplotlib`: A Matplotlib template for SuperMongo style, software.
48. **Li J.**, Melchior P. M., Hahn C., Huang S., Population-Level Inference for Galaxy Properties from Broadband Photometry, ICML ML4Astro workshop (2023).
49. Han J. J., Dey A., Price-Whelan A. M., Najita J., Schlafly E. F. et al. (including **Li J.**), NANCY: Next-generation All-sky Near-infrared Community surveY, white paper for the Roman Core Community Survey, arXiv:2306.11784.

Observational Programs

30 orbits on the Hubble Space Telescope, 15 nights on the 6.5 m Magellan telescopes, 3 nights on the 8 m Subaru telescope, 2 nights on the 8 m Gemini telescope, 10 nights on the 3.5 m WIYN telescope, plus various other observational programs.

PI:

Hubble Space Telescope Cycle 33: WFC3/UVIS imaging (30 orbits)	2025
Magellan Telescope: 11 dark nights, imaging with Baade/IMACS	2024–2025
Subaru Telescope: 25 hours dark time, imaging with Hyper-Suprime Camera	2025A
Green Bank Telescope: 84.5 hours	2025A, 2026A
Gemini-North telescope: 28.5 hours dark time imaging with GMOS	2024B, 2025B, 2026A
WIYN telescope: 10 dark nights, imaging with ODI	2024A, 2024B
Magellan Telescope: 1 dark night with Clay/IFUM	2025A
Magellan Telescope: 1 dark night, spectroscopy with Clay/LDSS3	2022B

Co-I:

Roman Cycle 1: Establishing the Roman Population II Distance Ladder (Analysis Proposal)	2026
JWST Cycle 5: Quenching in Isolation: JWST Star-Formation Histories of Field Dwarf Galaxies	2026

JWST Cycle 5: UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon	2026
JWST Cycle 4: J-Virgo: A JWST Treasury Survey of the Virgo Cluster	2025
Magellan Telescope: 4 dark nights, spectroscopy with Baade/MagE	2024–2025
HST Cycle 32: Galactic Underdogs: Assessing the True Satellite Galaxy Population in the Local Volume	2024
HST Cycle 32: Fulfilling the UV Legacy of the Hubble and Webb Deep Public Frontier Field	2024
Magellan Telescope: 3 dark nights, spectroscopy with Clay/IFUM	2023–2024
Merian Survey using the Dark Energy Camera on the Blanco telescope: 12 nights	2021–2024
Shane 3-m Telescope, UCO Lick Observatory: 2 nights observing	2019

References

Prof. Risa Wechsler	Stanford University
Prof. Jenny Greene	Princeton University
Prof. Marla Geha	Yale University
Prof. Shany Danieli	Tel Aviv University
Prof. Song Huang	Tsinghua University
Prof. Yingjie Peng	Kavli Institute for Astronomy and Astrophysics, Peking University

Software

- `smplotlib`: A Matplotlib template for the SuperMongo style.
  It has 130+ stars on GitHub and is used in 30+ published papers.
- `RoSESim`: Roman SEmi-resolved galaxy Simulator (Li+ in prep)
- `mrf`: Multi-Resolution Filtering – a method for isolating faint extended emission in Dragonfly data and other low-resolution images (van Dokkum et al. 2020)
- `bib2cv`: Convert a BibTeX file of publications into LaTeX entries for an academic CV.
- More work can be found on my GitHub: `@AstroJacobLi`

Collaboration, Teaching, Advising, Service, and Outreach

• Core member of the Merian Survey collaboration	2021 – now
• Contributor to LSST System Integration, Test, and Commissioning	2024 – now
• Co-advising Princeton graduate student: Lucas Mandacarú Guerra (1 paper submitted)	2025 – 2026
• Co-advising Princeton undergraduate students: Sufia Birmingham, Vivek Vijayakumar	
• Reviewer for ApJ, ApJL, AJ, A&A	2023 – now
• Course Designer of AST 207 (Observational Astronomy) at Princeton	2025 Spring
• Teaching Assistant for AST 303 (Research Methods in Astrophysics) at Princeton	2022 Fall
• Organizer of the Survey Science Discussion at Princeton	2022 – 2025
• Organizer of tea time for colloquium speakers at Princeton	2021 – 2022
• President of Peking University's Youth Astronomy Society	2017 – 2018

- Problem designer for the 2018 International Olympiad on Astronomy and Astrophysics (IOAA)
- Invited to a popular Chinese science TV show "Voice" (开讲啦) and talked about astronomy outreach

Talks and Presentations

Invited Talk, KICP, Probe Combination for Dark Matter Physics in the Era of Large Surveys — Sept 2026

Thesis Talk, AAS 247, Phoenix, Arizona — Jan 2026

Invited Talk, NASA Hyperwall at AAS 247, Phoenix, Arizona — Jan 2026

Revealing the Smallest Galaxies in the Nearby Universe with Roman

Invited Talk, Amateur Astronomers Association of Princeton (AAAP), Princeton, New Jersey — Dec 2026

Invited Talk, KICP Seminar, University of Chicago — Nov 2025

Talk, Observer Group Meeting, Northwestern University — Oct 2025

Invited Talk, Galaxy Seminar, University of Michigan — Oct 2025

Talk, ITC Luncheon, Harvard University — Oct 2025

Invited Talk, Journal Club, Rutgers University — Sept 2025

Talk, Near field science with the Roman High Latitude Wide Area Survey, Caltech — Aug 2025

How to characterize semi-resolved dwarf galaxies in Roman HLWAS

Invited Talk, KIPAC Tea Talk, Stanford University, California — Aug 2025

Dwarfs: Near and Far

Tea Talk, Carnegie Observatories, California — Aug 2025

Talk, Santa Cruz Galaxy Workshop, Santa Cruz, California — Aug 2025

ELVES-Dwarf Survey: Probing Satellite Population of Isolated Dwarf Galaxies in the Local Volume

Talk, Galactic Frontiers II: Dwarf Galaxies in the Local Volume and Beyond, Dartmouth — June 2025

ELVES-Dwarf Survey: Probing Satellite Population of Isolated Dwarf Galaxies in the Local Volume

Invited Talk, DREAMS collaboration meeting, New York City — May 2025

Invited Talk, Yale Galaxy Lunch — May 2025

Talk, Magellan Science Meeting, Washington D.C. — May 2025

Talk, Yale's Astro $\times$ Data Science seminar — Nov 2024

PopSED: Population-level inference for galaxy properties from broadband photometry

Talk, "Small Galaxies, Cosmic Questions. II", Durham, UK — Aug 2024

Poster, "Dwarf Galaxies, Star Clusters, and Streams in the LSST Era", Chicago, Illinois — July 2024

Hedgehog: An Isolated Quiescent Dwarf Galaxy at 2.4 Mpc

Talk, the 243rd meeting of the American Astronomical Society, New Orleans, Louisiana — Jan 2024

The Merian Survey: Mapping Classical Dwarf Galaxies at $z = 0.05 - 0.1$ with HSC-SSP + Blanco/DECam

Poster, the 243rd meeting of the American Astronomical Society, New Orleans, Louisiana — Jan 2024

PopSED: Population-level inference for galaxy properties from broadband photometry

Invited Talk, Tsinghua DoA Machine Learning Seminar — Nov 2023

Poster, ML for Astrophysics, International Conference on Machine Learning, Hawaii — July 2023

Talk, "Galactic Frontiers: Dwarf Galaxies in the Local Volume and Beyond", New York City — July 2023

Talk, Kavli Institute for Astronomy and Astrophysics, Peking University, Beijing June 2023

Ultra-puffy galaxies among satellites of Milky Way analogs: from definition to environmental quenching

Talk, Department of Astronomy, Tsinghua University, Beijing June 2023